TOÁN ỨNG DỤNG CHO KỸ SƯ

COURSE BOOK

ERWIN KREYSZIG, *Advanced engineering mathematics* .

CONTENTS

1. Complex Number (part D; chapter 13)

2. Linear Algebra (part B; chapter 7,8)

3. Differential Equation(part A; chapter 1,2,6)

4. Fourier Analysis (part C; chapter 11)

5. Optimization (part F; chapter 22)

Complex Number

$$z = (x, y).$$

x is called the **real part** and y the **imaginary part** of z, written

$$x = \operatorname{Re} z, \qquad y = \operatorname{Im} z.$$

In practice, complex numbers $z = (x, y)$ are written

$$i^2 = -1$$

$$z = x + iy$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2).$$

$$z = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\,\frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}.$$

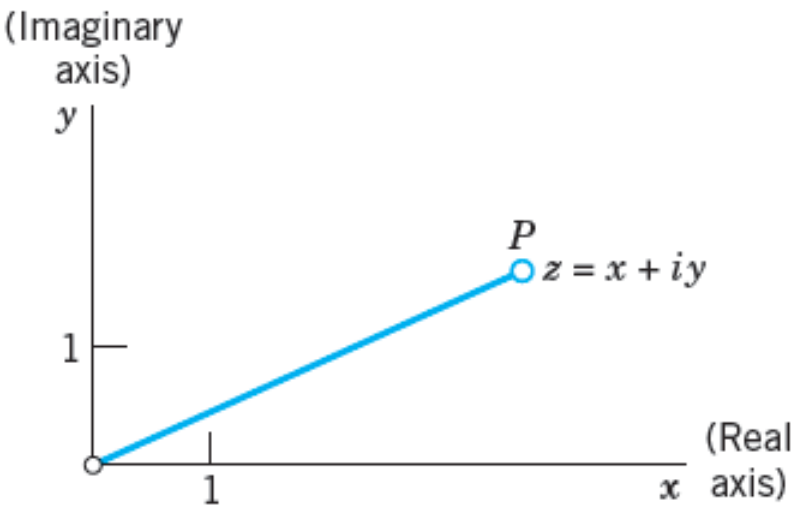


Fig. 318. The complex plane

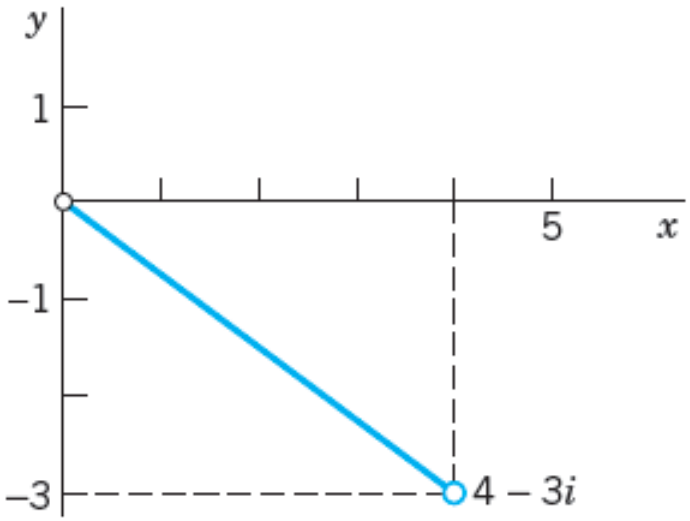


Fig. 319. The number $4 - 3i$ in the complex plane

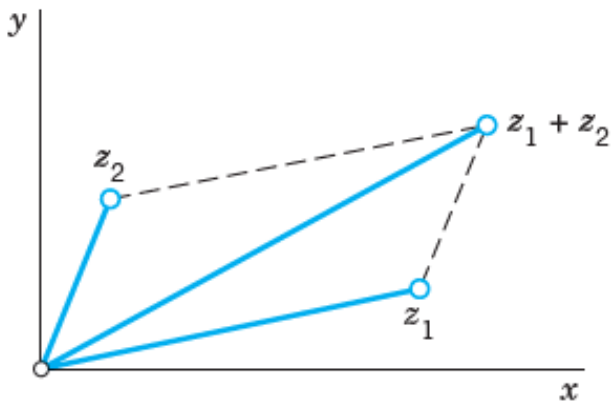


Fig. 320. Addition of complex numbers

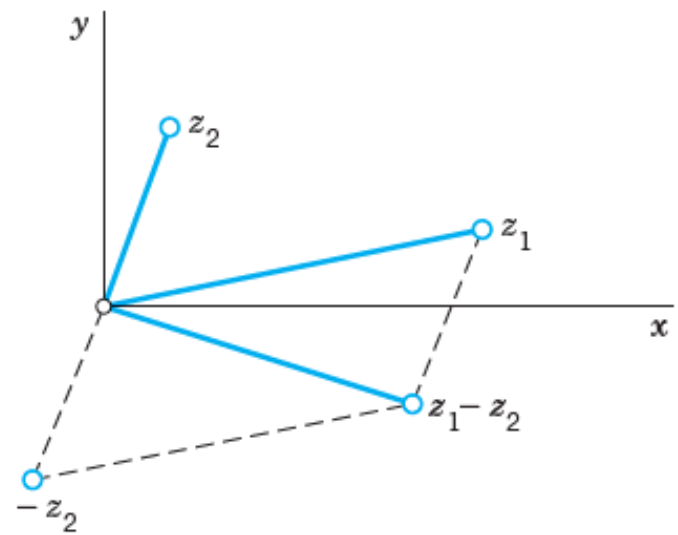
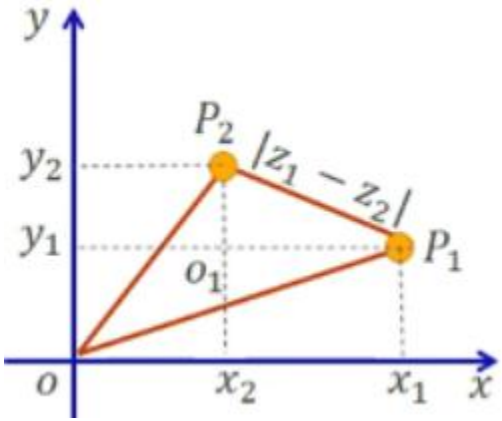


Fig. 321. Subtraction of complex numbers

Khoảng cách giữa 2 số phức

$$P_1P_2 = \sqrt{{o_1P_1}^2 + {o_1P_2}^2} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = |z_1 - z_2|$$



Complex Conjugate Numbers

The complex conjugate $\bar{z}$ of a complex number $z = x + iy$ is defined by

$$\bar{z} = x - iy.$$

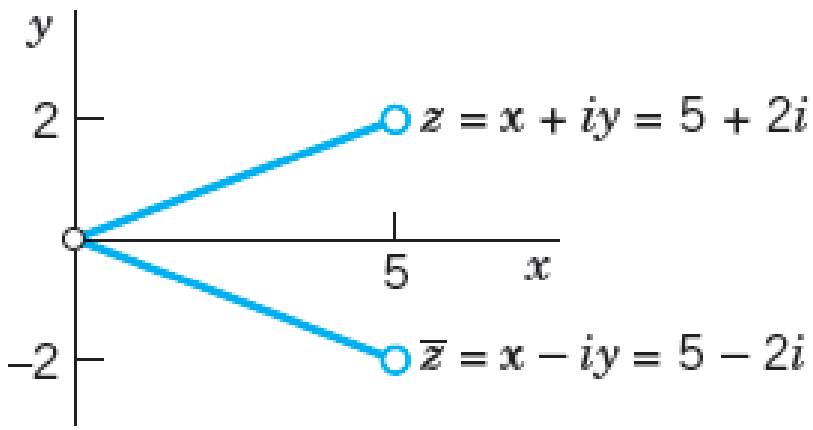


Fig. 322. Complex conjugate numbers

Polar Form of Complex Numbers.

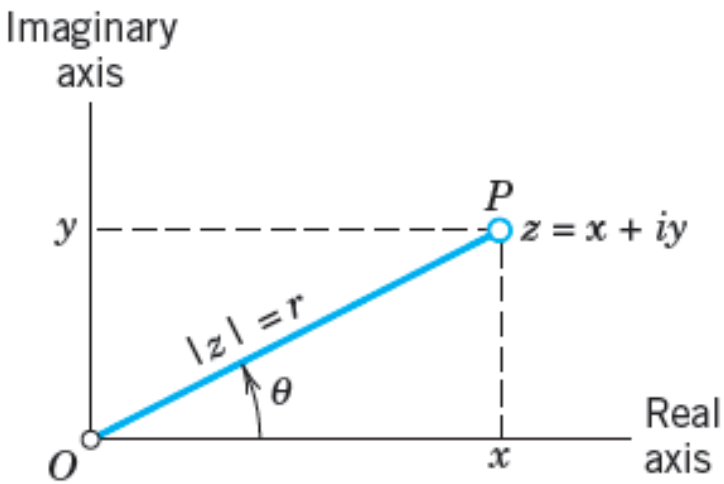


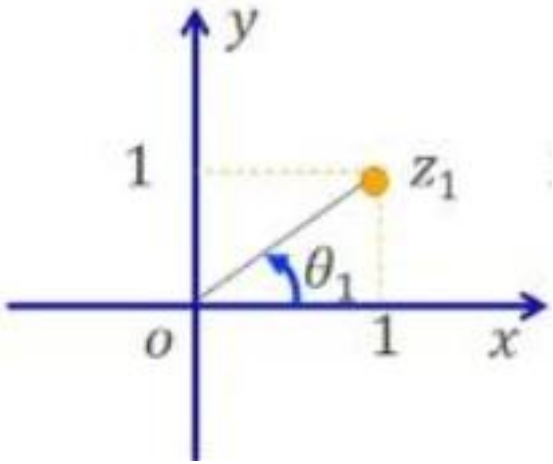
Fig. 323. Complex plane, polar form of a complex number

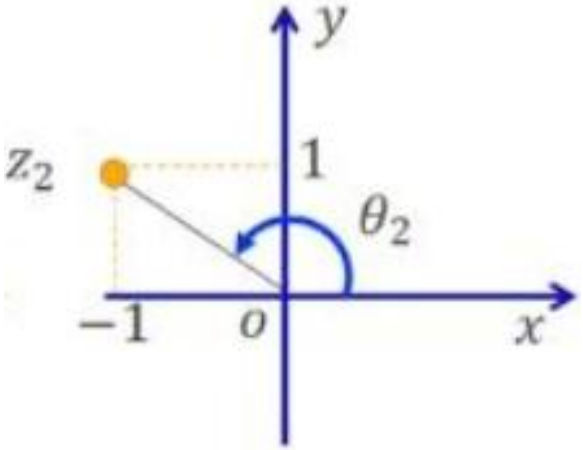
$$|z| = r = \sqrt{x^2 + y^2}$$

$$\tan\theta = \frac{y}{x}$$

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

$$z = r(\cos\theta + i\sin\theta).$$





- **Số phức bằng nhau**

$$z_1 = z_2 \Leftrightarrow \begin{cases} r_1 = r_2 \\ \theta_1 = \theta_2 + k2\pi \end{cases}$$

- **Nhân số phức**

$$z_1 z_2 = r_1 e^{j\theta_1} r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1+\theta_2)} = r_1 r_2 \angle \theta_1 + \theta_2$$

- **Chia số phức**

$$\frac{z_1}{z_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1-\theta_2)} = \frac{r_1}{r_2} \angle \theta_1 - \theta_2$$

$$z^n = r^n(\cos n\theta + i \sin n\theta).$$

$$\sqrt[n]{z} = \sqrt[n]{r}\left(\cos\frac{\theta + 2k\pi}{n} + i \sin\frac{\theta + 2k\pi}{n}\right)$$

$$k = 0, 1, \cdots, n - 1.$$

$$z^n = r^n(\cos n\theta + i \sin n\theta).$$

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = 1 - \sqrt{3}i$.

SOLUTION In Example 1 we saw that

$$z = 2\left[\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)\right].$$

Hence from (8) with $r = 2$, $\theta = -\pi/3$, and $n = 3$, we get

$$(1 - \sqrt{3}i)^3 = 2^3\left[\cos\left(3\left(-\frac{\pi}{3}\right)\right) + i \sin\left(3\left(-\frac{\pi}{3}\right)\right)\right]$$

$$= 8[\cos(-\pi) + i \sin(-\pi)] = -8.$$

DeMoivre's Formula When $z = \cos\theta + i \sin\theta$, we have $|z| = r = 1$ and

$$(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta.$$

EXAMPLE 4 Roots of a Complex Number

Find the three cube roots of $z = i$.

SOLUTION With $r = 1$, $\theta = \arg z = \pi/2$, the polar form of the given number is $z = \cos(\pi/2) + i\sin(\pi/2)$. From (10) with $n = 3$ we obtain

$$w_k = (1)^{1/3}\left[\cos\left(\frac{\pi/2 + 2k\pi}{3}\right) + i\sin\left(\frac{\pi/2 + 2k\pi}{3}\right)\right], \quad k = 0, 1, 2.$$

Hence, the three roots are

$$k = 0, \quad w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

≡

EXAMPLE 5 Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

SOLUTION In this case, $r = \sqrt{2}$ and $\theta = \arg z = \pi/4$. From (10) with $n = 4$, we obtain

$$w_k = (\sqrt{2})^{1/4}\left[\cos\left(\frac{\pi/4 + 2k\pi}{4}\right) + i\sin\left(\frac{\pi/4 + 2k\pi}{4}\right)\right], \quad k = 0, 1, 2, 3.$$

The roots, rounded to four decimal places, are

$$k = 0, \quad w_0 = (\sqrt{2})^{1/4}\left[\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right] = 1.0696 + 0.2127i$$

$$k = 1, \quad w_1 = (\sqrt{2})^{1/4}\left[\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right] = -0.2127 + 1.0696i$$

$$k = 2, \quad w_2 = (\sqrt{2})^{1/4}\left[\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right] = -1.0696 - 0.2127i$$

$$k = 3, \quad w_3 = (\sqrt{2})^{1/4}\left[\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right] = 0.2127 - 1.0696i.$$

PROBLEM SET 13.1

8–15 COMPLEX ARITHMETIC

Let $z_1 = -2 + 11i$, $z_2 = 2 - i$. Showing the details of your work, find, in the form $x + iy$:

8. $z_1 z_2$, $\overline{(z_1 z_2)}$

9. $\text{Re}\,(z_1^2)$, $(\text{Re}\, z_1)^2$

10. $\text{Re}\,(1/z_2^2)$, $1/\text{Re}\,(z_2^2)$

11. $(z_1 - z_2)^2/16$, $(z_1/4 - z_2/4)^2$

12. z_1/z_2, z_2/z_1

13. $(z_1 + z_2)(z_1 - z_2)$, $z_1^2 - z_2^2$

14. $\bar{z}_1/\bar{z}_2$, $\overline{(z_1/z_2)}$

15. $4\,(z_1 + z_2)/(z_1 - z_2)$

16–20

Let $z = x + iy$. Showing details, find, in terms of x and y:

16. $\text{Im}\,(1/z)$, $\text{Im}\,(1/z^2)$

17. $\text{Re}\, z^4 - (\text{Re}\, z^2)^2$

18. $\text{Re}\,[(1 + i)^{16} z^2]$

19. $\text{Re}\,(z/\bar{z})$, $\text{Im}\,(z/\bar{z})$

20. $\text{Im}\,(1/\bar{z}^2)$

PROBLEM SET 13.2

1–8 POLAR FORM

Represent in polar form and graph in the complex plane as in Fig. 325. Do these problems very carefully because polar forms will be needed frequently. Show the details.

1. $1 + i$

2. $-4 + 4i$

3. $2i, \quad -2i$

4. -5

5. $\dfrac{\sqrt{2} + i/3}{-\sqrt{8} - 2i/3}$

6. $\dfrac{\sqrt{3} - 10i}{-\frac{1}{2}\sqrt{3} + 5i}$

7. $1 + \frac{1}{2}\pi i$

8. $\dfrac{-4 + 19i}{2 + 5i}$

9–14 PRINCIPAL ARGUMENT

Determine the principal value of the argument and graph it as in Fig. 325.

9. $-1 + i$

10. $-5, \quad -5 - i, \quad -5 + i$

11. $3 \pm 4i$

12. $-\pi - \pi i$

13. $(1 + i)^{20}$

14. $-1 + 0.1i, \quad -1 - 0.1i$

15–18 CONVERSION TO $x + iy$

Graph in the complex plane and represent in the form $x + iy$:

15. $3\,(\cos \frac{1}{2}\pi - i \sin \frac{1}{2}\pi)$

16. $6\,(\cos \frac{1}{3}\pi + i \sin \frac{1}{3}\pi)$

17. $\sqrt{8}\,(\cos \frac{1}{4}\pi + i \sin \frac{1}{4}\pi)$

18. $\sqrt{50}\,(\cos \frac{3}{4}\pi + i \sin \frac{3}{4}\pi)$

21–27 ROOTS

Find and graph all roots in the complex plane.

21. $\sqrt[3]{1 + i}$

22. $\sqrt[3]{3 + 4i}$

23. $\sqrt[3]{216}$

24. $\sqrt[4]{-4}$

25. $\sqrt[4]{i}$

26. $\sqrt[8]{1}$

27. $\sqrt[5]{-1}$

28–31 EQUATIONS

Solve and graph the solutions. Show details.

28. $z^2 - (6 - 2i)z + 17 - 6i = 0$

29. $z^2 + z + 1 - i = 0$

30. $z^4 + 324 = 0$. Using the solutions, factor $z^4 + 324$ into quadratic factors with *real* coefficients.

31. $z^4 - 6iz^2 + 16 = 0$

Complex Function

$$w = f(z) = u(x, y) + iv(x, y).$$

w is complex, and we write $w = u + iv$, where u and v are the real and imaginary parts, respectively. Now w depends on $z = x + iy$. Hence u becomes a real function of x and y, and so does v.

EXAMPLE

Let $w = f(z) = z^2 + 3z$. Find u and v and calculate the value of f at $z = 1 + 3i$.

Solution. $u = \operatorname{Re} f(z) = x^2 - y^2 + 3x$ and $v = 2xy + 3y$. Also,

$$f(1 + 3i) = (1 + 3i)^2 + 3(1 + 3i) = 1 - 9 + 6i + 3 + 9i = -5 + 15i.$$

This shows that $u(1, 3) = -5$ and $v(1, 3) = 15$. Check this by using the expressions for u and v.

Complex Derivative

$$f'(z) = u_x + iv_x.$$

$$f'(z) = -iu_y + v_y.$$

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}.$$

The ***differentiation rules*** *are the same as in real calculus,* since their proofs are literally the same. Thus for any differentiable functions f and g and constant c we have

$$(cf)' = cf', \quad (f+g)' = f' + g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

as well as the chain rule and the power rule $(z^n)' = nz^{n-1}$ (n integer).

Cauchy–Riemann equations

$$f'(z) = u_x + iv_x \qquad f'(z) = -iu_y + v_y$$

Roughly, f is analytic in a domain D if and only if the first partial derivatives of u and v satisfy the two **Cauchy–Riemann equations**[4]

$$u_x = v_y, \qquad u_y = -v_x$$

EXAMPLE

Is $f(z) = u(x, y) + iv(x, y) = e^x(\cos y + i \sin y)$ analytic?

Solution. We have $u = e^x \cos y$, $v = e^x \sin y$ and by differentiation

$$u_x = e^x \cos y, \qquad v_y = e^x \cos y$$

$$u_y = -e^x \sin y, \qquad v_x = e^x \sin y.$$

We see that the Cauchy–Riemann equations are satisfied and conclude that $f(z)$ is analytic for all z.

Laplace's Equation. Harmonic Functions

If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then both u and v satisfy
Laplace's equation

$$\nabla^2 u = u_{xx} + u_{yy} = 0$$

$$\nabla^2 v = v_{xx} + v_{yy} = 0$$

u và v là hàm điều hòa.

u là hàm điều hòa liên hợp của v và ngược lại.

EXAMPLE

(a) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane.
(b) Find the harmonic conjugate function of u.

Solution.

we see that u satisfies Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0.$$

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 3x^2 - 3y^2 \qquad \text{and} \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 6xy + 5.$$

$$v(x, y) = 3x^2y - y^3 + h(x).$$

$$\frac{\partial v}{\partial x} = 6xy + h'(x).$$

$h'(x) = 5$, and so $h(x) = 5x + C$. $\qquad v(x, y) = 3x^2y - y^3 + 5x + C.$

EXAMPLE

Verify that $u = x^2 - y^2 - y$ is harmonic in the whole complex plane and find a harmonic conjugate function v of u.

Solution. $\nabla^2 u = 0$ by direct calculation. Now $u_x = 2x$ and $u_y = -2y - 1$. Hence because of the Cauchy–Riemann equations a conjugate v of u must satisfy

$$v_y = u_x = 2x, \qquad v_x = -u_y = 2y + 1.$$

Integrating the first equation with respect to y and differentiating the result with respect to x, we obtain

$$v = 2xy + h(x), \qquad v_x = 2y + \frac{dh}{dx}.$$

A comparison with the second equation shows that $dh/dx = 1$. This gives $h(x) = x + c$. Hence $v = 2xy + x + c$ (c any real constant) is the most general harmonic conjugate of the given u. The corresponding analytic function is

$$f(z) = u + iv = x^2 - y^2 - y + i(2xy + x + c) = z^2 + iz + ic.$$

PROBLEM SET 13.4

2–11 CAUCHY–RIEMANN EQUATIONS

Are the following functions analytic? Use (1) or (7).

2. $f(z) = iz\bar{z}$
3. $f(z) = e^{-2x}(\cos 2y - i \sin 2y)$
4. $f(z) = e^{x}(\cos y - i \sin y)$
5. $f(z) = \text{Re}\,(z^2) - i\,\text{Im}\,(z^2)$
6. $f(z) = 1/(z - z^5)$
7. $f(z) = i/z^8$
8. $f(z) = \text{Arg}\, 2\pi z$
9. $f(z) = 3\pi^2/(z^3 + 4\pi^2 z)$
10. $f(z) = \ln |z| + i\,\text{Arg}\, z$
11. $f(z) = \cos x \cosh y - i \sin x \sinh y$

12–19 HARMONIC FUNCTIONS

Are the following functions harmonic? If your answer is yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

12. $u = x^2 + y^2$
13. $u = xy$
14. $v = xy$
15. $u = x/(x^2 + y^2)$
16. $u = \sin x \cosh y$
17. $v = (2x + 1)y$
18. $u = x^3 - 3xy^2$
19. $v = e^{x} \sin 2y$
20. **Laplace's equation.** Give the details of the derivative of (9).

21–24

Determine a and b so that the given function is harmonic and find a harmonic conjugate.

21. $u = e^{\pi x} \cos av$
22. $u = \cos ax \cosh 2y$
23. $u = ax^3 + bxy$
24. $u = \cosh ax \cos y$

Exponential Function

$$e^z = e^x(\cos y + i \sin y).$$

(4) $$e^z = e^x e^{iy}.$$

Furthermore, for $z = iy$ we have from (1) the so-called **Euler formula**

(5) $$e^{iy} = \cos y + i \sin y.$$ $$e^{iz} = \cos z + i \sin z$$

Hence the **polar form** of a complex number, $z = r(\cos\theta + i \sin\theta)$, may now be written

(6) $$z = re^{i\theta}.$$

From (5) we obtain

(7) $$e^{2\pi i} = 1$$

as well as the important formulas (verify!)

(8) $$e^{\pi i/2} = i, \qquad e^{\pi i} = -1, \qquad e^{-\pi i/2} = -i, \qquad e^{-\pi i} = -1.$$

Ví dụ Giải phương trình $e^z = 3 + 4j$

Giải

Biểu diễn số phức về dạng mũ

$$3 + 4j = 5\angle 0{,}927 = 5e^{j0{,}927}$$

$$e^z = e^x e^{jy} = 5e^{j0{,}927}$$

$e^x = 5$ và $y = 0{,}927 + 2k\pi$

$$z = x + jy = \ln 5 + j(0{,}927 + 2k\pi)$$

PROBLEM SET 13.5

1. e^z is entire. Prove this.

2–7 **Function Values.** Find e^z in the form $u + iv$ and $|e^z|$ if z equals

2. $3 + 4i$

3. $2\pi i(1 + i)$

4. $0.6 - 1.8i$

5. $2 + 3\pi i$

6. $11\pi i/2$

7. $\sqrt{2} + \frac{1}{2}\pi i$

8–13 **Polar Form.** Write in exponential form (6):

8. $\bar{z}$

9. $4 + 3i$

10. $\sqrt{i}$, $\sqrt{-i}$

11. -6.3

12. $1/(1 - z)$

13. $1 + i$

14–17 **Real and Imaginary Parts.** Find Re and Im of

14. $e^{-\pi z}$

15. $\exp(z^2)$

16. $e^{1/z}$

17. $\exp(z^3)$

19–22 **Equations.** Find all solutions and graph some of them in the complex plane.

19. $e^z = 1$

20. $e^z = 4 + 3i$

21. $e^z = 0$

22. $e^z = -2$

Trigonometric Function

$$\cos z = \tfrac{1}{2}(e^{iz} + e^{-iz}), \qquad \sin z = \frac{1}{2i}(e^{iz} - e^{-iz}).$$

(a) $\cos z = \cos x \cosh y - i \sin x \sinh y$

(b) $\sin z = \sin x \cosh y + i \cos x \sinh y$

$$\cosh y = \tfrac{1}{2}(e^{y} + e^{-y}), \qquad \sinh y = \tfrac{1}{2}(e^{y} - e^{-y});$$

Hyperbolic Functions

The complex **hyperbolic cosine** and **sine** are defined by the formulas

(11) $$\cosh z = \tfrac{1}{2}(e^{z} + e^{-z}), \qquad \sinh z = \tfrac{1}{2}(e^{z} - e^{-z}).$$

$$\cosh iz = \cos z, \qquad \sinh iz = i \sin z.$$

$$\cos iz = \cosh z, \qquad \sin iz = i \sinh z.$$

Ví dụ Giải phương trình $\cos z = 5$

Giải

$$\cos z = 5 \Leftrightarrow \frac{e^{jz}+e^{-jz}}{2} = 5$$

$$10e^{jz} = \left(e^{jz} + e^{-jz}\right)e^{jz}$$

$$\left(e^{jz}\right)^2 - 10e^{jz} + 1 = 0$$

$$e^{jz} = \begin{cases} 9{,}899 \\ 0{,}101 \end{cases} = e^{j(x+jy)} = e^{-y}e^{jx}$$

$$\begin{cases} e^{-y} = \begin{cases} 9{,}899 \\ 0{,}101 \end{cases} \\ x = 2k\pi \end{cases} \Leftrightarrow \begin{cases} y = \begin{cases} -2{,}292 \\ 2{,}292 \end{cases} \\ x = 2k\pi \end{cases} \Leftrightarrow \begin{cases} z = 2k\pi + j(-2{,}292) \\ z = 2k\pi + j2{,}292 \end{cases}$$

PROBLEM SET 13.6

1–4 FORMULAS FOR HYPERBOLIC FUNCTIONS

Show that

1. $\cosh z = \cosh x \cos y + i \sinh x \sin y$

   $\sinh z = \sinh x \cos y + i \cosh x \sin y.$

2. $\cosh (z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$

   $\sinh (z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2.$

3. $\cosh^2 z - \sinh^2 z = 1, \quad \cosh^2 z + \sinh^2 z = \cosh 2z$
4. **Entire Functions.** Prove that $\cos z$, $\sin z$, $\cosh z$, and $\sinh z$ are entire.
5. **Harmonic Functions.** Verify by differentiation that $\operatorname{Im} \cos z$ and $\operatorname{Re} \sin z$ are harmonic.

6–12 Function Values. Find, in the form $u + iv$,

6. $\sin 2\pi i$
7. $\cos i, \quad \sin i$
8. $\cos \pi i, \quad \cosh \pi i$
9. $\cosh (-1 + 2i), \quad \cos (-2 - i)$
10. $\sinh (3 + 4i), \quad \cosh (3 + 4i)$
11. $\sin \pi i, \quad \cos (\frac{1}{2}\pi - \pi i)$
12. $\cos \frac{1}{2}\pi i, \quad \cos [\frac{1}{2}\pi(1 + i)]$

Logarithm. General Power. Principal Value

$$\ln z = \ln r + i\theta \qquad (r = |z| > 0, \quad \theta = \arg z).$$

$$\ln z = \operatorname{Ln} z \pm 2n\pi i$$

$\operatorname{Ln} z$ (Ln with capital L) and is called the **principal value** of $\ln z$.

$$\operatorname{Ln} z = \ln |z| + i \operatorname{Arg} z$$

$$\ln 4 = 1.386294 \pm 2n\pi i \qquad \operatorname{Ln} 4 = 1.386294$$

$$\ln(-1) = \pm\pi i, \pm 3\pi i, \pm 5\pi i, \cdots \qquad \operatorname{Ln}(-1) = \pi i$$

$$\ln(-4) = 1.386294 \pm (2n+1)\pi i \qquad \operatorname{Ln}(-4) = 1.386294 + \pi i$$

$$\ln i = \pi i/2, -3\pi/2, 5\pi i/2, \cdots \qquad \operatorname{Ln} i = \pi i/2$$

$$\ln 4i = 1.386294 + \pi i/2 \pm 2n\pi i \qquad \operatorname{Ln} 4i = 1.386294 + \pi i/2$$

General powers of a complex number $z = x + iy$ are defined by

(7) $$z^c = e^{c \ln z} \qquad a^z = e^{z \ln a}.$$

PROBLEM SET 13.7

COMPLEX NATURAL LOGARITHM *ln z*

5–11 **Principal Value Ln z.** Find Ln z when z equals

5. -11
6. $4 + 4i$
7. $4 - 4i$
8. $1 \pm i$
9. $0.6 + 0.8i$
10. $-15 \pm 0.1i$
11. ei

12–16 **All Values of ln z.** Find all values and graph some of them in the complex plane.

12. $\ln e$
13. $\ln 1$
14. $\ln(-7)$
15. $\ln(e^i)$
16. $\ln(4 + 3i)$
17. Show that the set of values of $\ln(i^2)$ differs from the set of values of $2 \ln i$.

18–21 **Equations.** Solve for z.

18. $\ln z = -\pi i/2$
19. $\ln z = 4 - 3i$
20. $\ln z = e - \pi i$
21. $\ln z = 0.6 + 0.4i$

22–28 **General Powers.** Find the principal value. Show details.

22. $(2i)^{2i}$
23. $(1 + i)^{1-i}$
24. $(1 - i)^{1+i}$
25. $(-3)^{3-i}$
26. $(i)^{i/2}$
27. $(-1)^{2-i}$
28. $(3 + 4i)^{1/3}$

11–16 **Complex Numbers.** Find, in the form $x + iy$, showing details,

11. $(2 + 3i)^2$

12. $(1 - i)^{10}$

13. $1/(4 + 3i)$

14. $\sqrt{i}$

15. $(1 + i)/(1 - i)$

16. $e^{\pi i/2}, \quad e^{-\pi i/2}$

17–20 **Polar Form.** Represent in polar form, with the principal argument.

17. $-4 - 4i$

18. $12 + i, \quad 12 - i$

19. $-15i$

20. $0.6 + 0.8i$

21–24 **Roots.** Find and graph all values of:

21. $\sqrt{81}$

22. $\sqrt{-32i}$

23. $\sqrt[4]{-1}$

24. $\sqrt[3]{1}$

25–30 **Analytic Functions.** Find $f(z) = u(x, y) + iv(x, y)$ with u or v as given. Check by the Cauchy–Riemann equations for analyticity.

25 $u = xy$

26. $v = y/(x^2 + y^2)$

27. $v = -e^{-2x} \sin 2y$

28. $u = \cos 3x \cosh 3y$

29. $u = \exp(-(x^2 - y^2)/2) \cos xy$

30. $v = \cos 2x \sinh 2y$

31–35 **Special Function Values.** Find the value of:

31. $\cos(3 - i)$

32. $\operatorname{Ln}(0.6 + 0.8i)$

33. $\tan i$

34. $\sinh(1 + \pi i), \quad \sin(1 + \pi i)$

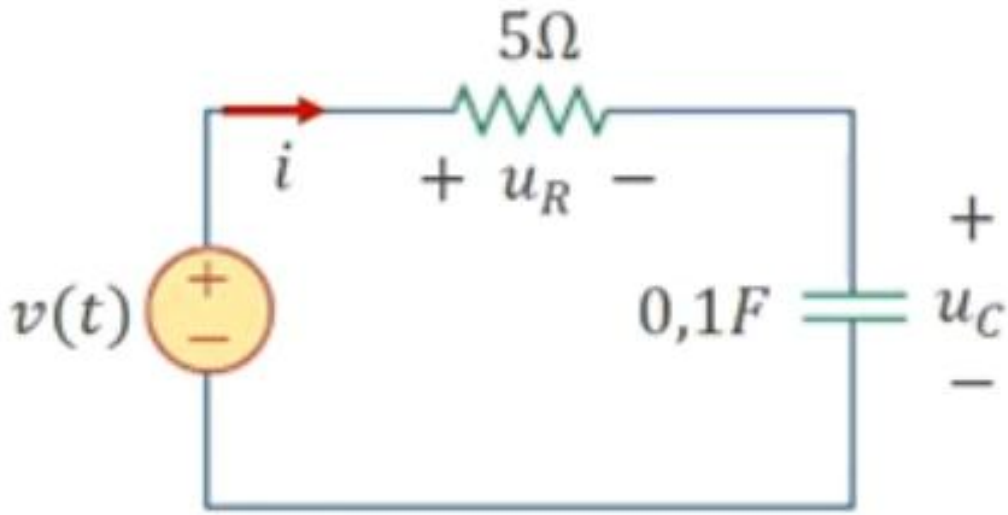
35. $\cosh(\pi + \pi i)$

Ứng dụng số phức trong kỹ thuật điện

Hàm điều hòa trong miền thời gian	Biểu diễn sang số phức
$u = U_m \cos(\omega t + \varphi)$	$U = U_m \angle \varphi$
$U_m \sin(\omega t + \varphi)$	$-jU_m \angle \varphi$
u'	$j\omega U_m \angle \varphi$
$\int u dt$	$\frac{U_m \angle \varphi}{j\omega}$

Quan hệ u và i của R, L, C	Biểu diễn sang số phức
$u = Ri$	$U = RI$
$u = Li'$	$U = Lj\omega I$
$u = \frac{1}{C}\int i dt$	$U = \frac{I}{Cj\omega}$

Ví dụ



$v(t) = 10\cos 4t$

- Viết phương trình $k2$ và biến đổi sang phương trình số phức.
- Tìm I và viết biểu thức $i(\mathrm{t})$.

Giải

- Phương trình $k2$

$$u_R + u_C = 10\cos 4t$$

$$5i + \frac{1}{0,1}\int idt = 10\cos 4t$$

- Phương trình số phức

$$5I + \frac{I}{0,1j4} = 10\angle 0$$

- Tìm I

$$I = \frac{10}{5 - 2,5j} = 1,79\angle 26,6^0 A$$

- Viết $i(t)$

$$i(t) = 1,79\cos(4t + 26,6^0)A$$